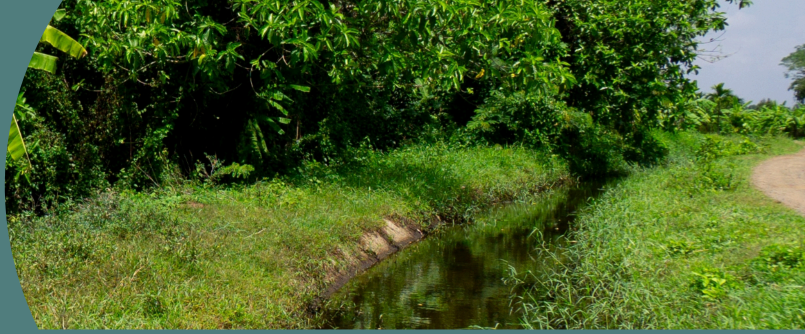
PUNJAB
DRAINAGE CHANNELS



Sector: Agriculture
Hazard adressed: Flood
Targeted assets: Rice and Cotton

Description: This measure involves construction and rehabilitation of drainage channels to safely convey excess floodwater and storm runoff away from vulnerable agricultural and settlement areas. The intervention reduces flood inundation, minimizes waterlogging, improves drainage efficiency, and protects crops, infrastructure, and livelihoods from flood-related damages.

Priority Districts: Sialkot, Jhang, Toba Tek Singh, Khanewal, Vehari, Bahawalnagar, Bahawalpur, Lodhran, Rajanpur, DG Khan, Muzaffargarh, Multan, Rahim Yar Khan.

TECHNICAL SPECIFICATIONS/DIMENSION

Unit of implementation: Kilometer (km) of drainage channel.

Scale assumptions: Applicable in flood-prone agricultural districts and low-lying areas vulnerable to drainage congestion and seasonal flooding.

Key design parameters: Drainage channel width and depth, hydraulic flow capacity, embankment stabilization, sediment management, culverts, and outfall structures designed according to local hydrological conditions.

Time horizon: 2030 – 2050.

IMPACT

Risk reduction level: High.

Time to effectiveness: Medium-term following completion of channel construction and operationalization.

Expected impact: Reduced flood inundation and waterlogging, improved drainage efficiency, protection of crops and infrastructure, and enhanced resilience of agricultural systems against flood hazards.

IMPLEMENTATION CONSIDERATIONS

Enabling conditions: Hydrological assessments, engineering design expertise, institutional coordination, financing support, and land availability.

Required institutions: Punjab Irrigation Department, Agriculture Department Punjab, PDMA Punjab, Local Government Department, and development partners.

Barriers: Land acquisition constraints, sedimentation, drainage blockage, weak maintenance systems, and limited long-term financing.

COST ASSUMPTIONS

Unit cost: PKR 26 million per km (~ USD 92,750 per km).

Capital Expenditure (CAPEX):
USD 10.7 million per year. Includes excavation, embankment construction, culverts, drainage structures, and associated civil works.

Operational Expenditure (OPEX):
USD 3.5 million per year. Includes desilting, vegetation removal, structural repairs, and routine maintenance.

CO-BENEFITS & TRADE-OFFS

Environmental: Reduced standing water and waterlogging, improved soil conditions, and reduced erosion risks.

Social: Improved safety and livelihood protection for flood-prone rural communities.

Economic: Reduced crop and infrastructure losses due to flooding and improved agricultural productivity.

Potential risks: High capital investment requirements, sediment accumulation, encroachment issues, and maintenance challenges during extreme flood events.

BENEFICIARIES

Primary beneficiaries: Farmers, irrigation departments, local governments, rural communities, and disaster management authorities.

Distributional aspects: Particularly beneficial for vulnerable agricultural communities located in flood-prone and poorly drained areas.